FM Host (fm_host) - flow chart

Analog FM repeater with an EchoLink link. Talks to the Modem Host on :18000+N-1 and to fm_gateway on :18700+N-1. Not available on the hotspot modems; the link needs an OpenMT (sample) modem.

main() - start-up

Start
fm_host -m N [-d] [-x]

Refuse hotspot modem types. Read FM settings (access mode, CTCSS, hang, CW ID, link levels, DTMF) from the database

Connect to the Modem Host, register MODE FM

Start fmClientThread (receive), fmTxThread (transmit) and serverThread (EchoLink gateway)

Service loop: fmServiceOnce() until stopped

Modem -> repeater and EchoLink (receive)

Modem Host
TCP :18000+N-1

packets

fmClientThread: read a packet from the Modem Host

Packet type?

command

Duplex or simplex flag from the Modem Host

FMDA, FMST, FMEO, FMRS

MMDVM firmware packets: fmFirmwareFrame(). The firmware repeats; the host only follows the call for the dashboard

samples

fmRxSamples(): discriminator samples at 24 kS/s. DC blocker, CTCSS detector, noise squelch, voice band filter into the delay ring

gateway connected

Decimate to 8 kS/s. DTMF decoder (Goertzel): a digit goes to the gateway as ELDT and its tones are muted. fmLinkRx(): audio through an 85 ms delay to the gateway as ELAU while linked; ELRX marks squelch open and closed

fmRxAccess(): CTCSS and squelch combined by the access mode

Signal?

opens, closes

fmPostEvent(): call start or end queued for the service loop

ELRX

fm_gateway
TCP :18700+N-1

tone drops

The latest audio is muted so the squelch tail is not repeated

Repeater and EchoLink -> modem (transmit)

Delay ring (receiver)
Link ring (EchoLink)

audio

fmTxThread: every 4 ms (one 96 sample frame, absolute clock) calls fmTxTick()

Transmitter state?

signal opens or link audio

Take the modem (SETMODE FM). Wait for the Modem Host to confirm FM mode, then key up

signal and link gone

Hold the carrier for the hang time, then stop

keyed

fmBuildFrame(): the delayed audio, the link audio, the CTCSS tone when one is set, and the CW ID while it runs

from the gateway

ELAU audio from the linked station: 8 to 24 kS/s, pre-buffered (linkPrebuf), mixed with the repeater audio. ELPT says when the station starts and stops talking

fmSendSamples(): one 96 sample frame sent to the Modem Host as SAMP

End: the EOT marker frame is sent and the modem is released (SETIDLE)

Modem Host
TCP :18000+N-1

Timers, dashboard commands and the database

fmServiceOnce(): call events written to the database, modem mode checked for the transmitter

Dashboard commands: updateConfFM reloads the settings; reflLinkFM (link,<node or callsign> / unlink) is sent to the gateway as CONN or DISC and answered by its CONN, NACK or DISC. gatewayThread serves the gateway socket

MMDVM modem: fmSendFirmwareParams() sends the FM settings at start-up, and the call timeout is watched

MariaDB
config, calls, history, dashboard commands, errors

Key

step · decision · start / thread · outside / database / network · modem hardware · queue / buffer

With the plain MMDVM modem the firmware does the CTCSS and repeater work itself and there is no EchoLink link; with the OpenMT modems this host does it in software. Errors go to the error log with reportError().